DDI-Net 3.0

Explainable drug interaction intelligence

Drug–Drug Interaction Research Report

Warfarin + Acetylsalicylic acid

DRUGS ANALYSED

A **Warfarin**
DB00682 · approved

B **Acetylsalicylic acid**
DB00945 · approved, vet_approved

Report ID	DDI-20261018-092726-A26BB655
Generated	18 October 2026, 09:27:26 (UTC, UTC+00:00)
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Model	BIO-GINE M4 · bd45f84e3c1b2c33 · seed 0
API report version	3.0
Analysis produced by the service	2026-10-18T09:27:23+00:00
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RESEARCH USE ONLY

DDI-Net is a research prototype — not a medical device and not clinical decision support. The absence of a documented interaction is not evidence of safety. The BIO-GINE score is a research signal of resemblance to documented pairs, not a probability of clinical harm.

1 Executive summary

Documented in a database

Interaction documented in a database or reference set; no severity grade is shown in this report

A database describes an interaction; it gives no severity.

This is the kind of documentation found, not a level of danger.

INTERACTION STATUS (OUTCOME ENGINE)

Documented interaction found

Sources name specific consequences or changes in effect for this pair. What they are, and on what basis, is below.

What could happen?

No specific clinical outcome established from available evidence. This does not mean the combination is safe.

- **Stronger anticoagulant (blood-thinning) effect** — change in effect; documented; severity not established. [1]

Seriousness: Severity not established from available evidence.

IN BRIEF

Documentation found — FDA label documents: 0 · distinct excerpts: 0 · DrugBank records: 1. Mechanism hypotheses: 2. BIO-GINE calibrated score: 0.76 — a research signal, not a risk.

FOUR KINDS OF INFORMATION IN THIS REPORT

Documented evidence

FDA labels: 0 (0 excerpts); DrugBank records: 1

A label or database record that names both drugs.

Mechanism hypotheses

hypotheses: 2; shared Reactome pathways: 11

Assembled from each drug's documented roles separately; not tested for the pair.

Machine-learning prediction

calibrated score 0.76 · regime S3

BIO-GINE's resemblance to documented pairs; not a probability of harm.

Missing or unavailable evidence

FDA labels (openFDA): unavailable

What could not be checked, or is absent from the sources.

What the official sources document

- DrugBank (TDC export) holds 1 interaction record(s) for this pair. A record gives the interaction type only — no severity, and no note of what it is based on. [1]

What the quoted evidence directly says

- In the sources' own words (shortened where marked "…"; full text in section 3). Only what these sentences state is documented; the report adds no interpretation to them.
- DrugBank, standard sentence; direction not verified
  "One of the two drugs may increase the anticoagulant activities of the other drug." [1]
- Direction: the drug order in the DrugBank export is not verified for this record type, so the report does not say which drug affects which.

What is only a hypothesis

- Acetylsalicylic acid → Warfarin: induces a protein that handles the other drug (CYP2C19). H1

- Warfarin → Acetylsalicylic acid: induces a protein that handles the other drug (CYP2C9). H2
- Assembled from what databases record about each drug separately; not tested for this pair. The arrow is the direction of the hypothesis, not a documented finding.
- Shared Reactome pathways: 11 (background, not a cause of interaction).

What BIO-GINE adds

- Calibrated score 0.76 (regime S3): the pair strongly resembles pairs documented as interacting. The model was not trained on this pair's record. Since documentation already exists, the score adds no new evidence that the drugs interact; it shows whether the model, on its own, would rank the pair with documented ones. M
- Measured quality on the S3 test pairs (neither drug seen in training, as here): AUPRC 0.737 ± 0.015. That is an average over the test set, not the reliability of this one prediction.
- One drug of this pair was in the validation set, whose pairs were used to choose the model's settings and to fit the calibration, so the test figures describe this pair only approximately.
- The score says nothing about mechanism, direction, severity or what to do, and it is not a probability of harm (more in section 5).

What remains uncertain

- Source “FDA labels (openFDA)”: unavailable. Its data is not part of this report.
- For 1 of 1 DrugBank records the drug order in the export is not verified, so the report does not say which drug affects which.
- Databases do not record how strongly a drug inhibits or induces a protein, or how much of a drug's handling that protein does.
- The service uses one model checkpoint (seed 0) and provides neither a per-pair confidence interval nor an ensemble estimate. Section 5 gives the quality measured on the held-out test instead — an average over many pairs, not confidence in this pair.
- No source gives a severity, and none is assigned here.
- Doses, duration, age, kidney and liver function, other medicines and individual factors are not taken into account.

Terms such as S2, AUPRC or CYP450 are explained in the glossary (section 9).

2 Drug information

FIELD	A · WARFARIN	B · ACETYLSALICYLIC ACID
International name (INN)	warfarin	acetylsalicylic acid
Russian name	варфарин (transliteration)	ацетилсалициловая кислота (-transliteration)
DrugBank ID	DB00682	DB00945
DrugBank groups	approved	approved, vet_approved
Type	small molecule	small molecule
Query as entered	DB00682	DB00945
Matched on	—	—
Protein annotations (targets / enzymes / transporters / carriers)	2 / 6 / 0 / 2	19 / 4 / 3 / 0
Model split (seed 0)	val	val
Protein data available to the model	yes	yes
DrugBank page	https://go.drugbank.com/drugs/DB00682	https://go.drugbank.com/drugs/DB00945

Identifiers: DrugBank. INN: DrugCentral, joined on InChIKey (no value means no match). Russian names: rule-based transliteration — an interface label, not sourced data.

3 Documented interactions

Evidence is grouped by source document: [n] is a document, [n.k] one distinct excerpt of it; the explanation refers to these labels. Several excerpts of one label are one source, not separate confirmations. The English text is the source's own words (an excerpt or DrugBank's standard sentence).

FDA label documents: 0 · distinct excerpts: 0 · DrugBank records: 1

Source documents

#	DOCUMENT	IDENTIFIER	VERSION / DATE	EXCERPTS	SECTIONS
[1]	DrugBank (TDC export)	DB00682|DB00945|Y=6	PyTDC 1.1.15, tdc.multi_pred.DDI(name='DrugBank')	1	—

[1] DrugBank (TDC export)

Retrieved 2026-08-24

1 interaction record(s); DrugBank records are never merged

DrugBank DDI records via Therapeutics Data Commons · PyTDC 1.1.15, tdc.multi_pred.DDI(name='DrugBank')

Database · DrugBank | direction not verified | Source's words · DrugBank template, direction-neutral

“One of the two drugs may increase the anticoagulant activities of the other drug.”

direction not verified: the report does not say which drug affects which

DOCUMENT PROVENANCE

URL	https://tdcommons.ai/multi_pred_tasks/ddi/
Record ID	DB00682|DB00945|Y=6
Retrieved	2026-08-24
Licence	TDC: CC BY 4.0; underlying DrugBank content: DrugBank non-commercial licence
Severity as stated by the source	not stated by the source

QUALIFIERS

- DrugBank's standard sentence for this interaction type, not a quotation from a study or a label.
- This export carries no severity grade. None is implied.
- The export does not state the basis of each record (clinical study, label text, or inference from pharmacological properties).
- The export's drug order for this interaction type cannot be checked, and for checkable types it proved unreliable (reversed in about 90% of records for types 47 and 73), so the statement is shown without naming which drug is affected.

Clinical outcomes (as the sources state them)

Consequences the sources state for this pair, each with its quotation, document, section and extraction rule. Seriousness only as a source states it.

No specific clinical outcome established from available evidence. This does not mean the combination is safe.

OUTCOME	KIND / STATUS	SERIOUSNESS	SOURCES
Stronger anticoagulant (blood-thinning) effect	change in effect documented	severity not established	[1]
Possible lower exposure of Warfarin (hypothesis) research only	change in exposure hypothesis	not applicable	–
Possible lower exposure of Acetylsalicylic acid (hypothesis) research only	change in exposure hypothesis	not applicable	–

change in effect · documented · severity not established

Stronger anticoagulant (blood-thinning) effect

The source describes a stronger (or altered) pharmacological effect of one of the drugs. It does not state that a particular harm follows.
Severity not established from available evidence. The source does not say which drug is affected (or its order is unverified); the report does not name one.
DrugBank standard sentence; the source does not state what the record is based on. Source documents: 1.
`outcome:DB00682+DB00945:bleeding_coagulation:anticoagulant_increase`

1. DrugBank DDI records via Therapeutics Data Commons · 2026-08-24

"#Drug1 may increase the anticoagulant activities of #Drug2."

`drugbank_type_map/v1:type=6 · anticoagulant · DB00682|DB00945|Y=6`
DrugBank gives no severity and does not state the basis of the record.

- A change in effect is described, not that harm occurs.
- The available evidence does not establish a mechanism for this outcome.

change in exposure · hypothesis · not applicable

Possible lower exposure of Warfarin (hypothesis) · Warfarin

Assembled from each drug's separate roles; not documented for this pair.
Not applicable: there is no documented outcome. The hypothesis's direction comes from the protein roles.
Only the drugs' separate protein roles (annotations). This is a hypothesis. Source documents: 0.
`outcome:DB00682+DB00945:exposure_decrease:hypothesis:DB00682`

1. DrugBank / ChEMBL drug–protein annotations

"CYP2C19 (enzyme): roles A — inhibitor, substrate; roles B — inducer"

`annotation_chain/v1:inducer_substrate · inducer_substrate · mech:P33261:enzyme:inducer_substrate:DB00945`

- The strength of inhibition and the protein's share of the work are unknown.

change in exposure · hypothesis · not applicable

Possible lower exposure of Acetylsalicylic acid (hypothesis) · Acetylsalicylic acid

Assembled from each drug's separate roles; not documented for this pair.
Not applicable: there is no documented outcome. The hypothesis's direction comes from the protein roles.
Only the drugs' separate protein roles (annotations). This is a hypothesis. Source documents: 0.
`outcome:DB00682+DB00945:exposure_decrease:hypothesis:DB00945`

1. DrugBank / ChEMBL drug–protein annotations

"CYP2C9 (enzyme): roles A — inducer, substrate; roles B — substrate"

`annotation_chain/v1:inducer_substrate · inducer_substrate · mech:P11712:enzyme:inducer_substrate:DB00682`

- The strength of inhibition and the protein's share of the work are unknown.

Sources checked

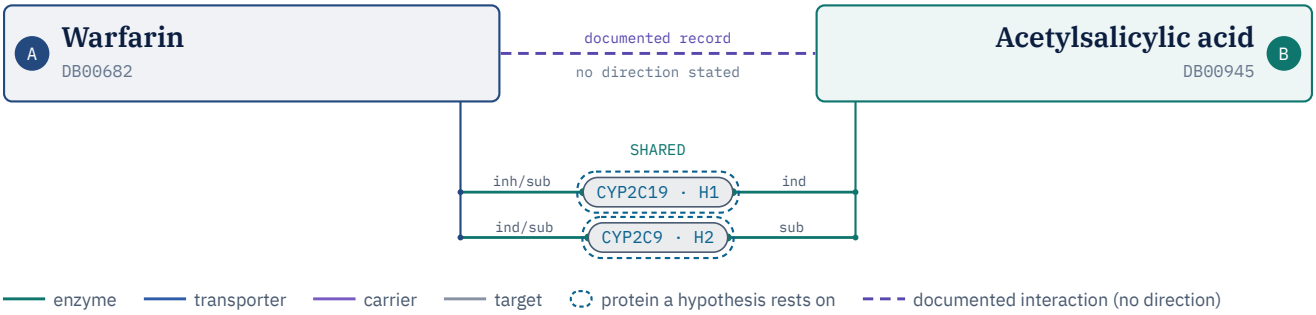
SOURCE	STATUS	DETAIL
DrugBank (TDC export, local)	checked	1 record(s) in the local DrugBank export (TDC, retrieved 2026-08-24).
FDA labels (openFDA)	unavailable	openfda_label: recently unreachable; retrying after a cooldown

4 Biological mechanisms

Below are each drug's documented roles on their own (DrugBank, ChEMBL) and how they line up. These are mechanism hypotheses, not evidence of an interaction: the databases do not record strength of effect or how much of a drug's processing a protein handles.

Mechanism graph

Drawn from the report's mechanism block: proteins documented for both drugs. The full graph, including proteins of only one drug, is on the Mechanisms page.



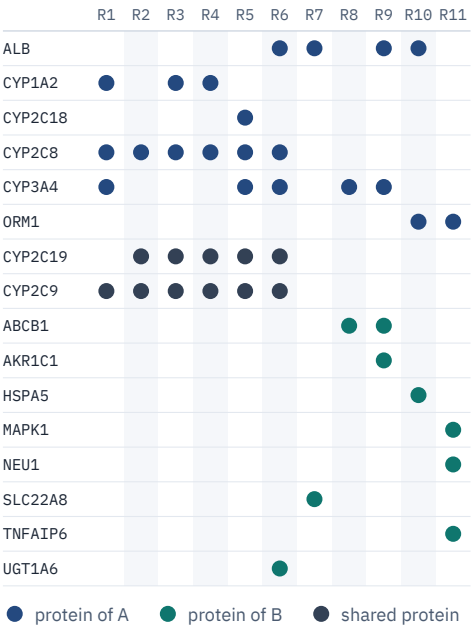
Roles on branches: sub — substrate, inh — inhibitor / inhibits, ind — inducer, act — activates, n/s — not stated.
The arc between the drugs is a documented interaction record. It has no arrowhead: the DrugBank export's drug order is not verified for this type.

Mechanism hypotheses

#	PROTEIN	CONFIGURATION	ROLES A / B	BASIS
H1	CYP2C19 Cytochrome P450 2C19 P33261 · enzyme	induces a protein that handles the other drug hypothesis: Acetylsalicylic acid → Warfarin	A: inhibitor, substrate B: inducer	DrugBank drug-protein annotation 5.1 (export 2025-01-02) PMID 19515014, 16162970, 16372822, 12621391
H2	CYP2C9 Cytochrome P450 2C9 P11712 · enzyme	induces a protein that handles the other drug hypothesis: Warfarin → Acetylsalicylic acid	A: inducer, substrate B: substrate	DrugBank drug-protein annotation 5.1 (export 2025-01-02) PMID 29283396, 16372822, 20735727, 21966608

Shared biological pathways (Reactome) · 11

11 of 11 shown: pathways through a shared protein first, then the most specific. Protein membership in Reactome pathways (a dot means the protein is in the pathway; colour shows whose protein it is: A, B or shared).



R	PATHWAY	REACTOME	PROTEINS	VIA A	VIA B
R1	Biosynthesis of maresin-like SPMs	https://reactome.org/content/detail/R-HSA-9027307	6	CYP1A2, CYP2C8, CYP2C9, CYP3A4	CYP2C9
R2	CYP2E1 reactions	https://reactome.org/content/detail/R-HSA-211999	8	CYP2C19, CYP2C8, CYP2C9	CYP2C19, CYP2C9
R3	Synthesis of (16-20)-hydroxyeicos-atetraenoic acids (HETE)	https://reactome.org/content/detail/R-HSA-2142816	8	CYP1A2, CYP2C19, CYP2C8, CYP2C9	CYP2C19, CYP2C9
R4	Synthesis of epoxy (EET) and dihydroxyeicosatrienoic acids (DHET)	https://reactome.org/content/detail/R-HSA-2142670	8	CYP1A2, CYP2C19, CYP2C8, CYP2C9	CYP2C19, CYP2C9
R5	Xenobiotics	https://reactome.org/content/detail/R-HSA-211981	16	CYP2C18, CYP2C19, CYP2C8, CYP2C9, CYP3A4	CYP2C19, CYP2C9
R6	Aspirin ADME	https://reactome.org/content/detail/R-HSA-9749641	25	ALB, CYP2C19, CYP2C8, CYP2C9, CYP3A4	CYP2C19, CYP2C9, UGT1A6
R7	Ciprofloxacin ADME	https://reactome.org/content/detail/R-HSA-9793528	5	ALB	SLC22A8
R8	Atorvastatin ADME	https://reactome.org/content/detail/R-HSA-9754706	8	CYP3A4	ABCB1
R9	Prednisone ADME	https://reactome.org/content/detail/R-HSA-9757110	10	ALB, CYP3A4	ABCB1, AKR1C1
R10	Platelet degranulation	https://reactome.org/content/detail/R-HSA-114608	44	ALB, ORM1	HSPA5
R11	Neutrophil degranulation	https://reactome.org/content/detail/R-HSA-6798695	154	ORM1	MAPK1, NEU1, TNFAIP6

From mechanism to outcome

Each link is marked: directly documented, verified annotation, or plausible but unconfirmed. A shared protein or pathway alone is never a link.

directly documented — verified biological annotation — plausible but unconfirmed ····· unsupported or unavailable



Acetylsalicylic acid induces CYP2C19, which processes Warfarin (annotations). The exposure change is a hypothesis.
verified biological annotation: Acetylsalicylic acid is annotated as an inducer of CYP2C19.
verified biological annotation: Warfarin is annotated as a substrate of CYP2C19.
plausible but unconfirmed: A change in exposure is plausible but not confirmed for this pair.



Warfarin induces CYP2C9, which processes Acetylsalicylic acid (annotations). The exposure change is a hypothesis.
verified biological annotation: Warfarin is annotated as an inducer of CYP2C9.
verified biological annotation: Acetylsalicylic acid is annotated as a substrate of CYP2C9.
plausible but unconfirmed: A change in exposure is plausible but not confirmed for this pair.

Proteins in the hypotheses: family background (project text, not a statement about this pair)

CYP2C19 — An enzyme of the cytochrome P450 family. These work mainly in the liver and gut and break down many medicines so the body can clear them.

CYP2C9 — An enzyme of the cytochrome P450 family. These work mainly in the liver and gut and break down many medicines so the body can clear them.

5 BIO-GINE research analysis

CALIBRATED SCORE	RAW SCORE	REGIME	RELIABILITY FOR THIS PAIR
0.76	0.9998	S3	like the held-out pairs of the frozen test

Numbers are rounded; exact values are in Appendix A and the companion JSON.

BIO-GINE is a research model that looks for patterns resembling documented interactions. Its score is not a probability of harm or an individual's risk, and it does not affect the status or the outcomes above.

Model and checkpoint	BIO-GINE M4 · bd45f84e3c1b2c33 · seed 0
Regime	S3 — neither drug was in the training set
Splits A / B	val / val
Pair's record seen in training	no
Protein data available to the model	A: yes · B: yes

What the two numbers mean

- Raw score (0.9998): the network's direct output, a number between 0 and 1. Near the ends of the scale the values bunch up: differences in the fourth or fifth decimal place mean nothing beyond “very high” (or “very low”).
- Calibrated score (0.76): the same output, rescaled by “temperature scaling”. The rescaling was fitted on validation pairs of which half were documented interactions and half random pairs. It pulls over-confident values back from the edges of the scale but does not change the order of pairs. That is why the two numbers differ.
- Neither number is the probability that the combination harms anyone. The model only learned to tell documented pairs from random ones. The calibrated score could be read as a share of documented pairs only in a half-and-half mix like the one it was fitted on; real drug combinations are not such a mix, and a documented interaction says nothing about how serious it is.
- What they can tell: how strongly the model groups this pair with pairs already documented as interacting. That is useful for ranking pairs for research.
- What they cannot tell: whether an interaction happens in a particular person, how strong or serious it would be, which drug affects which, or what anyone should do.

Measured model quality (held-out test set, not this pair)

METRIC	MEAN ± SD	SEEDS	TEST SET	SOURCE
AUPRC	0.737 ± 0.015	5	S3 (both drugs unseen in training)	paper/FACTS.json models.M4_s3 (v2_final_s3_posthoc.csv @ frozen tag)
AUROC	0.735 ± 0.018	5	S3, both annotated	reports/v3_coverage_reliability.json

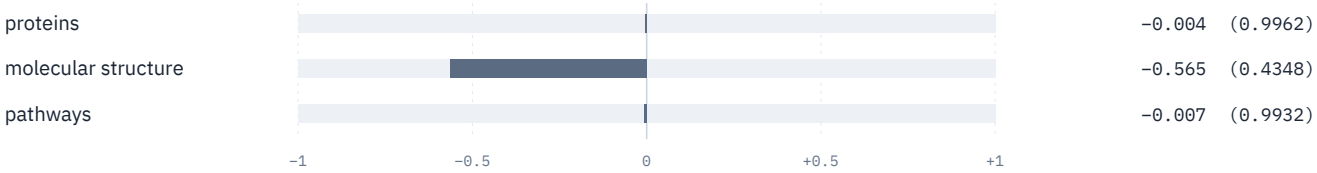
- This pair's regime is S3; the figures are for the 7,758 S3 test pairs, in which the model saw neither drug during training.
- “both annotated”: only test pairs whose two drugs both have protein data (79,712 pairs of the pooled test).
- AUPRC and AUROC describe the model's quality over thousands of test pairs; “±” is the spread between 5 training runs. They do not say how likely this particular prediction is to be right. The service gives no uncertainty for a single pair: it uses one trained copy of the model (seed 0).
- One drug of this pair was in the validation set, whose pairs were used to choose the model's settings and to fit the calibration, so the test figures describe this pair only approximately.

Uncertainty

The service uses one model checkpoint (seed 0) and provides neither a per-pair confidence interval nor an ensemble estimate. Section 5 gives the quality measured on the held-out test instead — an average over many pairs, not confidence in this pair.

What the model leans on (one kind of data zeroed)

Change in the raw score when one kind of data is removed for both drugs. A minus sign means the score drops without that data. In brackets: the raw score without this data.



Raw score: 0.9998

The raw score is near the edge of the scale (0 or 1); changes are compressed there, and reliance can look smaller than it is.

This shows which data the model leans on for this pair, not why the drugs interact: the model never saw zeroed inputs in training.

Method check: reproduces the frozen V2 analysis on 84,690 test rows, max |Δ| = 6.1e-6.

`Method: branch zeroed before the frozen fusion layer for both drugs; mask unchanged (as scripts/49_v2_interpretability.py at the frozen tag)`

Notes from the service

- One drug was in the validation set: the model was not trained on it, but it was used to choose the configuration and the calibration.
- The calibrated score was fitted on validation pairs with equal numbers of documented and randomly drawn pairs. It is not a probability of harm, nor how often interactions happen in practice.

6 Structured explanation from the service

The text below was written by the service with deterministic templates over the retrieved records (no language model) and checked by it for advice wording. Labels point to records: [n] source document, [n.k] one of its excerpts, H hypothesis, M model, R Reactome pathway. The text below comes from the service; its terms (S2, AUPRC, AUROC) are explained in the glossary (section 9).

What is known

Source — DrugBank describes this combination as: “One of the two drugs may increase the anticoagulant activities of the other drug.” [1]

Caveat — For this record type the drug order in the DrugBank export proved unreliable (we measured it), so the text does not say which drug affects which. [1]

Why this may happen

Hypothesis — acetylsalicylic acid is recorded as inducing CYP2C19, and warfarin as something CYP2C19 processes. With more CYP2C19 available, warfarin may be broken down faster than usual. H1

Hypothesis — warfarin is recorded as inducing CYP2C9, and acetylsalicylic acid as something CYP2C9 processes. With more CYP2C9 available, acetylsalicylic acid may be broken down faster than usual. H2

Caveat — The databases record neither how strongly a drug inhibits or induces a protein nor how much of a drug's handling that protein does. These are hypotheses, not a conclusion that an interaction will happen. H1 H2

Annotation — Proteins of both drugs take part in 11 shared biological process(es) in Reactome — for example “Biosynthesis of maresin-like SPMs”. A shared process is background, not a cause of interaction. R

What the AI found

Model — BIO-GINE finds this pair strongly resembles pairs already recorded as interacting. This is resemblance to recorded pairs, not an estimate of harm. M

Model — For this pair the model's score depends most on the molecular structure: without it the score moves by 0.56. This says what the model leans on, not why the drugs interact. M

Caveat — One drug was in the validation set: the model was not trained on it, but it was used to choose the configuration and the calibration. M

Caveat — The calibrated score was fitted on validation pairs with equal numbers of documented and randomly drawn pairs. It is not a probability of harm, nor how often interactions happen in practice. M

How certain is this?

Source — Documented: 1 DrugBank record(s). Several excerpts of one label are one source, not separate confirmations, and a DrugBank record may itself rest on label text. This is the most reliable part of the page. [1]

Caveat — No severity is assigned: the DrugBank export has none, and for labels only the section that names the drug is shown.

Hypothesis — The 2 mechanism hypothesis(es) are assembled from each drug's documented roles separately. None was tested for this pair. H1 H2

Model — Model quality on held-out pairs where the model saw neither drug in training (S3, as here): AUPRC 0.737 ± 0.015 (spread across 5 training runs). That is an average over many pairs, not confidence in this particular pair. M

What is missing

Missing — The source “FDA labels (openFDA)” could not be checked right now. Its data is not part of this answer.

Caveat — Doses, duration, age, kidney and liver function, other medicines and individual factors are not taken into account — everything that decides whether an interaction shows up and how strongly.

Caveat — Decisions about medicines belong to a doctor or pharmacist. This page cannot replace them.

Method: deterministic templates over retrieved records; no language model

7 Sources and reproducibility

Evidence records

#	SOURCE	IDENTIFIER	RETRIEVED	URL
[1]	DrugBank DDI records via Therapeutics Data Commons · PyTDC 1.1.15, tdc.multi_pred.DDI(name='DrugBank')	DB00682|DB00945|Y=6	2026-08-24	https://tdcommons.ai/multi_pred_tasks/ddi/

Protein annotations the hypotheses rest on

DRUG	PROTEIN	ROLES	SOURCE	RETRIEVED	PUBMED
Warfarin	CYP2C19 P33261	inhibitor, substrate raw action: substrate|inhibitor	DrugBank drug-protein annotation · 5.1 (export 2025-01-02)	2026-08-28	19515014, 16162970, 16372822
Acetylsalicylic acid	CYP2C19 P33261	inducer raw action: inducer	DrugBank drug-protein annotation · 5.1 (export 2025-01-02)	2026-08-28	12621391
Warfarin	CYP2C9 P11712	inducer, substrate raw action: substrate|inducer	DrugBank drug-protein annotation · 5.1 (export 2025-01-02)	2026-08-28	29283396, 16372822, 20735727
Acetylsalicylic acid	CYP2C9 P11712	substrate raw action: substrate	DrugBank drug-protein annotation · 5.1 (export 2025-01-02)	2026-08-28	21966608

PubMed: https://pubmed.ncbi.nlm.nih.gov/<PMID>/ · UniProt: https://www.uniprot.org/uniprotkb/<accession>/entry

Provenance and parameters

Endpoint (report)	`http://127.0.0.1:8765/api/v3/report`
Request body (report)	`{"drug_a":"DB00682","drug_b":"DB00945","include_labels":true}`
Response received by the browser (report)	2026-10-18T09:27:23.827Z
Report generated by the service (generated_at)	2026-10-18T09:27:23+00:00
API report version	3.0
Explanation method	deterministic templates over retrieved records; no language model
Model	BIO-GINE M4 · bd45f84e3c1b2c33 · seed 0
Frozen research tag	`v2-final-github-safe-2026-09-03 · 92c481eeaba8faff991ced850e1c4de418ea31b0`
Site	http://127.0.0.1:4173
Site build commit	not recorded
PDF generator	DDI-Net web, in-browser export (@react-pdf/renderer)
Generated	18 October 2026, 09:27:26 (UTC, UTC+00:00)
Generated (UTC)	2026-10-18T09:27:26.507Z
Language	English
SHA-256 of the response data	`a26bb655b2ad2f8c80c4809b3d4aa9e7010ec7f0871f6b04fe3bdf1e451b7346` Hash of the service response(s) as canonical JSON (keys sorted, serialised with JSON.stringify). The same hash is in the companion JSON file, available from the “JSON” button next to “Export PDF”.

Retrieval times are given per record (retrieved_at) as returned by the service: the export date for local extracts, the request date for labels.

8 Limitations

Core limitations

- DDI-Net is a research tool, not clinical decision support and not a medical device. Decisions about medicines belong to a doctor or pharmacist.
- Unlabelled pairs are not confirmed safe. A missing record means only that the sources checked do not contain one.
- BIO-GINE scores are not calibrated probabilities of clinical harm: temperature scaling was fitted on validation pairs at 50% prevalence against randomly drawn unlabelled pairs.
- The drug order in the DrugBank (TDC) export was verified by measurement for only two interaction types; for the others it is either meaningless (symmetric types) or unverified, so directional claims (“A raises the level of B”) are not made.
- External medical evidence (FDA labels, RxNorm) may be incomplete or unavailable at query time; its status is listed under “Sources checked”.

Limitations reported by the service for this report

- Only the 1,705 drugs of the research dataset; 1,497 of them have an international name.
- DrugBank records carry no severity and do not state their basis. For many record types the drug order in the export is unreliable (measured).
- Labels are searched by drug name. Statements about whole drug classes (“strong CYP3A4 inhibitors”) are not recognised, so no match does not mean the label is silent.
- Mechanism hypotheses are assembled from each drug's roles separately; strength of effect and the protein's share of the work are unknown.
- The BIO-GINE score is a research signal of resemblance to documented pairs. It is not a probability of harm.

> RESEARCH USE ONLY
> This is a research prototype, not a medical device. It does not replace a doctor or pharmacist and cannot say whether a combination is safe for a particular person.

9 Glossary

Plain explanations of the terms used in this report.

Source evidence vs model prediction	Source evidence is text that an official label or a database actually contains about these two drugs. A model prediction is a number BIO-GINE computes from patterns in the data it was trained on. The first records what is known; the second is a research guess.
FDA label (SPL)	A medicine's official prescribing information in the United States, published by the FDA as Structured Product Labeling. One label is one document with its own set ID, divided into sections such as Warnings or Drug Interactions.
Document and excerpt	An excerpt is one sentence quoted from a document. Several excerpts of one label are still one source and do not confirm each other. When a label repeats a sentence in two sections, it is shown once.
DrugBank (TDC export)	A large database of drugs and known interactions; DDI-Net uses the public copy distributed by Therapeutics Data Commons. A record states only an interaction type, in a standard sentence, with no severity and no note of what it is based on.
Direction	Which drug affects which. For most interaction types the drug order in the DrugBank export is unreliable, so the report names no direction from DrugBank. Label sentences are quoted as written; no direction is inferred from the mere fact that one drug's label names the other.
Cytochrome P450 (CYP450)	A family of enzymes, mostly in the liver (for example CYP2C9, CYP3A4), that break down many medicines. If one drug slows an enzyme that breaks down another, the second may stay in the body longer.
Substrate, inhibitor, inducer	A substrate is a drug a protein processes. An inhibitor slows the protein down. An inducer makes the body produce more of it.
Mechanism hypothesis	A possible biological explanation assembled from what databases record about each drug separately (for example: drug X inhibits an enzyme that processes drug Y). It was not tested for this pair and says nothing about the size of any effect.
Reactome pathway	A curated chain of reactions in the cell. Two drugs sharing a pathway is background information, not a cause of interaction.
Raw and calibrated score	The raw score is the model's direct output between 0 and 1; the calibrated score is the same output rescaled on balanced validation pairs. Both put pairs in the same order. Neither is a probability of harm.
Regimes S1, S2, S3	Whether the model saw the drugs during training. S1: both drugs were in training (quality not measured). S2: one drug was not in training. S3: neither was — the hardest test.
Held-out test; “pooled”	Pairs set aside in advance and never used for training; quality is measured on them. “Pooled” means S2 and S3 test pairs measured together (84,690 pairs, 7,758 of them S3).

AUROC	The chance that the model ranks a randomly chosen documented pair above a randomly chosen pair from the comparison set (random pairs without a record). 0.5 is guessing; 1 is a perfect ranking.
AUPRC	Area under the precision–recall curve: how well the model finds documented pairs without raising many false alarms, over all thresholds at once. Half of the test pairs are documented, so guessing gives about 0.5; 1 is perfect.
± (spread between runs)	The model was trained 5 times, each from a different random starting point (seed). "±" is the standard deviation of the result between those runs. It is not an error bar for a single pair.

S1 Supplement S1. Evidence records as returned by the service

Every record of the API response, with its section, identifiers and retrieval time. Records that repeat one label's text map to the same excerpt; the full response is in the companion JSON file.

[1] `drugbank-tdc:DB00682>DB00945:t6`

Category	Database · DrugBank
Source	DrugBank DDI records via Therapeutics Data Commons · PyTDC 1.1.15, tdc.multi_pred.DDI(name='DrugBank')
Record ID	`DB00682|DB00945|Y=6`

Original identifiers	`drugbank_id_drug1_as_exported: DB00682`
	`drugbank_id_drug2_as_exported: DB00945`
	`tdc_interaction_type: 6`

URL	https://tdcommons.ai/multi_pred_tasks/ddi/
Retrieved	2026-08-24
DrugBank template	#Drug1 may increase the anticoagulant activities of #Drug2.
Direction	direction not verified: the report does not say which drug affects which
Licence	TDC: CC BY 4.0; underlying DrugBank content: DrugBank non-commercial licence

A Appendix A. Machine-readable summary (JSON)

Key fields as JSON for automated checking. The full service responses are in the companion JSON file (same SHA-256).


```
{
  "schema": "ddinet.pdf-summary/1",
  "report_id": "DDI-20261018-092726-A26BB655",
  "generated_at": "2026-10-18T09:27:26+00:00",
  "generated_at_utc": "2026-10-18T09:27:26.507Z",
  "time_zone": "UTC",
  "language": "en",
  "payload_sha256": "a26bb655b2ad2f8c80c4809b3d4aa9e7010ec7f0871f6b04fe3bdf1e451b7346",
  "inputs": [
    {"endpoint":"http://127.0.0.1:8765/api/v3/report","request":{"drug_a":"DB00682","drug_b":"DB00945","include_labels":true},
      "received_at":"2026-10-18T09:27:23.827Z"}
  ],
  "server_generated_at": "2026-10-18T09:27:23+00:00",
  "report_version": "3.0",
  "drugs": [
    {"role":"A","id":"DB00682","inn":"warfarin","name_ru":"варфарин"},
    {"role":"B","id":"DB00945","inn":"acetylsalicylic acid","name_ru":"ацетилсалициловая кислота"}
  ],
  "evidence_status": "documented_no_severity",
  "evidence_counts": {
    "labelDocuments": 0,
    "labelExcerpts": 0,
    "noEffectExcerpts": 0,
    "labelRecords": 0,
    "mergedRepeats": 0,
    "databaseRecords": 1,
    "referenceRecords": 0
  },
  "source_documents": [
```

```
    {"ref":"[1]","kind":"database","document_id":"drugbank_tdc","owner":null,"mentions":null,"excerpts":[{"ref":"[1]","category":
      "database_interaction_record","sections":[],"records":["drugbank-tdc:DB00682>DB00945:t6"]}]}
  ],
  "evidence": [
    {"ref":"[1]","id":"drugbank-tdc:DB00682>DB00945:t6","category":"database_interaction_record","source":
      "DrugBank DDI records via Therapeutics Data Commons","record_id":"DB00682|DB00945|Y=6","section":null,"also_in_sections":[],
      "direction":"not_verified","retrieved_at":"2026-08-24","url":"https://tdcommons.ai/multi_pred_tasks/ddi/"}
  ],
  "conflicts": 0,
  "hypotheses": [
    {"ref":"H1","gene":"CYP2C19","uniprot":"P33261","relation":"enzyme","pattern":"inducer_substrate","actor":"DB00945","affected":
      "DB00682"},
    {"ref":"H2","gene":"CYP2C9","uniprot":"P11712","relation":"enzyme","pattern":"inducer_substrate","actor":"DB00682","affected":
      "DB00945"}
  ],
  "shared_pathways_total": 11,
  "model": {
    "model": "BIO-GINE M4",
    "checkpoint": "bd45f84e3c1b2c33",
    "available": true,
    "calibrated_score": 0.7641836505184048,
    "raw_score": 0.9997894984380711,
    "regime": "S3",
    "reliability": "held_out_like",
    "abstain": false
  },
  "sources_checked": [
    {"source":"drugbank_tdc","status":"ok"},
    {"source":"openfda_label","status":"unavailable"}
  ],
  "outcomes": {
    "interaction_status": "documented_interaction",
    "seriousness": "documented_no_standard_severity",
    "records": [{"id":"outcome:DB00682+DB00945:bleeding_coagulation:anticoagulant_increase","category":"bleeding_coagulation",
      "effect":"anticoagulant_increase","level":"pharmacodynamic_effect","status":"documented","seriousness":
      "documented_no_standard_severity","research_only":false,"affected":null,"evidence":["drugbank-tdc:DB00682>DB00945:t6"],
      "methods":["drugbank_type_map/v1:type=6"]},{"id":"outcome:DB00682+DB00945:exposure_decrease:hypothesis:DB00682","category":
      "exposure_decrease","effect":"exposure_hypothesis","level":"pharmacokinetic_change","status":"hypothesized","seriousness":
      "not_applicable","research_only":true,"affected":"DB00682","evidence":["mech:P33261:enzyme:inducer_substrate:DB00945"],
      "methods":["annotation_chain/v1:inducer_substrate"]},{"id":"outcome:DB00682+DB00945:exposure_decrease:hypothesis:DB00945",
      "category":"exposure_decrease","effect":"exposure_hypothesis","level":"pharmacokinetic_change","status":"hypothesized",
      "seriousness":"not_applicable","research_only":true,"affected":"DB00945","evidence":[
      "mech:P11712:enzyme:inducer_substrate:DB00682"],"methods":["annotation_chain/v1:inducer_substrate"]}]
  }
}
```